A Multi-Axis Convolution Network for Small and High-Aspect-Ratio Object Detection in Remote Sensing

Feng Zhang, Jiaofeng Wang, Jie Mao, Chunru Dong, Qiang Hua, and Jie Zhu

***Abstract*—Remote sensing object detection (RSOD) is vital for tasks ranging from environmental monitoring to infrastructure analysis. However, due to the unique perspective and extensive spatial coverage of remote sensing images, feature attenuation and localization errors occur, posing significant challenges for reliable detection of small objects and high-aspect-ratio (HAR) objects in object detection tasks. To address these limitations, we propose the multi-axis convolution network (MACNet), a novel architecture designed to enhance feature representation for small objects and improve shape discrimination for HAR objects. MACNet introduces two innovative modules: the large-kernel dual-branch (LKDB) module, which utilizes large-kernel convolutions and parallel spatial and channel attention to enhance contextual understanding and mitigate feature attenuation of small objects, and the direction-aware convolution (DC) module, which employs dual orthogonal strip convolutions to capture distinctive shape patterns that are crucial for HAR objects. In addition, we propose a dynamically weighted KFIoU (DKFIoU) loss function to balance the contribution of center point loss and intersection over union (IoU) loss. DKFIoU adaptively weights the center point loss and a Gaussian-approximated IoU based on object size and aspect ratio, thereby improving the detection of HAR objects and prioritizing small objects by reducing localization errors. Comprehensive evaluations on three benchmark datasets demonstrate that MACNet significantly outperforms current state-of-the-art methods, particularly for small objects and HAR objects. Ablation studies provide valuable insights into the efficacy of our proposed components (https://github.com/ww66671/mac_net).**

***Index Terms*—High-aspect-ratio (HAR) object detection, object detection, remote sensing, small object detection.**

I. Introduction

OBJECT detection plays a vital role in remote sensing applications, such as urban planning [1], [2], maritime surveillance [3], environmental monitoring [4], and agriculture [5], [6]. Although significant progress has been made, reliably detecting small objects and high-aspect-ratio (HAR) objects remains a great challenge, hindering the potential of these applications. As illustrated in Fig. 1, it is difficult to distinguish the ships and small vehicles due to their limited spatial resolution inherent in remote sensing images. In this work, we follow the common definition of small objects as those occupying less than 32×32 pixels in the image space [8], emphasizing that their small scale is manifested in images. Since small objects only occupy a few pixels, most existing deep learning-based object detection models inevitably avoid the feature attenuation, which means the receptive fields of the small objects in the feature space will progressively shrink as the network goes deeper, which hinders the extraction of discriminative features, leading to a high false detection rate.

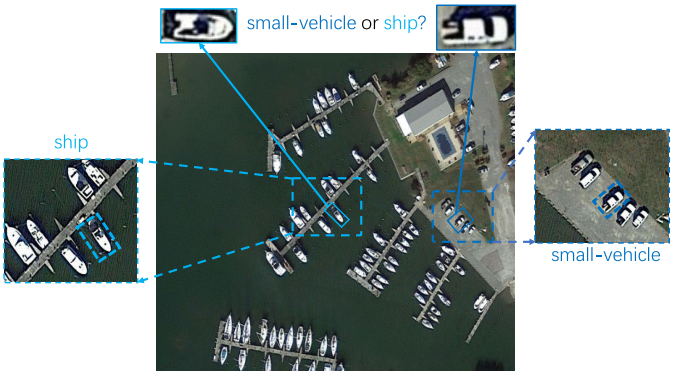


Fig. 1. Accurate identification of objects with limited spatial dimensions, exemplified by small vehicles and ships, poses a significant challenge.

Recent work on small object detection has pursued several directions to mitigate feature attenuation; however, challenges remain in balancing context and noise suppression. YOLO-based improvements [9], [10] can reduce attenuation of small objects but often struggle to suppress noise in complex backgrounds since they emphasize local features. Enhancements to feature pyramid networks (FPNs) [11], [12], [13] can alleviate redundant feature generation by carefully designed feature fusion mechanisms to preserve details, yet the upsampling smoothing can dilute fine-grained information that is critical to small objects. Large kernel convolutions [14], [15] extend receptive fields to capture richer context, but their

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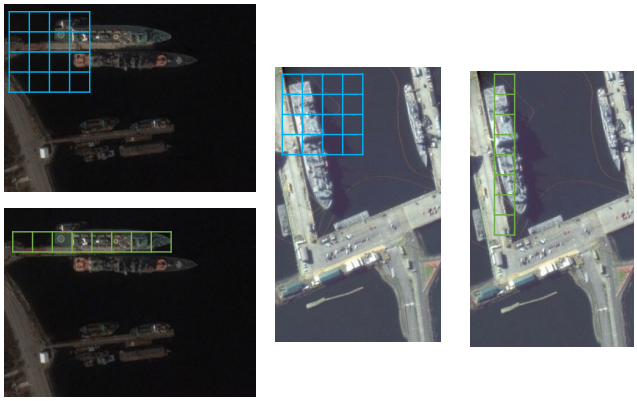


Fig. 2. Detection of HAR objects benefits from the specialized information captured by different kernel shapes. Square and strip convolutional kernels extract disparate features, as visualized by the blue and green bounding boxes, respectively. Notably, strip convolutional kernels exhibit reduced inclusion of irrelevant information compared to square kernels, making them more suitable for accurate identification of HAR objects.

fixed, position-invariant weights impede adaptive filtering of irrelevant background regions. Attention-based methods [16], [17] offer dynamic feature weighting; however, they may overemphasize salient local cues at the expense of broader contextual information. In summary, these approaches struggle to achieve an optimal balance between contextual information and background noise suppression for small-object detection.

To address the tradeoff, we propose the large-kernel dual-branch (LKDB) module. Unlike methods that merely stack large kernel convolutions or employ generic attention mechanisms, LKDB introduces a dual-branch structure that enables synergistic feature adaptive calibration and noise suppression while leveraging large kernels to enlarge the receptive field for contextual modeling. To further prevent background interference, the LKDB introduces a dual-branch spatial–channel attention mechanism, where the spatial branch captures long-range contextual dependencies, and the channel branch adaptively emphasizes discriminative features. This coordinated mechanism enhances small-object representations while maintaining structural simplicity, effectively suppressing background noise in complex remote sensing scenes.

Another persistent challenge in remote sensing object detection (RSOD) lies in the accurate identification of HAR objects. These objects are often characterized by linear or strip-shaped features with a length-to-width ratio significantly greater than one, manifested as elongated shapes exhibiting distinct directionality. While expanding the spatial receptive field using large kernels can benefit small object detection, it inevitably introduces background noise for HAR objects. As shown in Fig. 2, employing the square kernels will include irrelevant regions when processing HAR objects. Existing studies [18], [19] utilize cascaded strip convolutions to model the directional dependencies and enhance shape features of elongated objects. However, the sequential application of strip convolutions ($1 \times N$ followed by $N \times 1$ or vice versa) hinders the network from extracting the orientation-specific features, resulting in a loss of crucial directional information.

To fundamentally address this directional bias, we design a novel direction-aware convolution (DC) module. The DC module employs parallel orthogonal strip convolutions to adaptively fuse multidirectional features. Unlike sequential feature extraction paradigms, such a parallel architecture not only eliminates inherent directional bias but also suppresses background noise introduced by standard square convolutions. By enabling simultaneous and equivalent capture of multidirectional features, it facilitates more robust representations of HAR objects in complex and cluttered remote sensing scenes.

While intersection over union (IoU) loss is widely used for bounding box regression [20], its inherent sensitivity to positional and rotational deviations poses significant challenges for small objects and HAR objects. As illustrated in Fig. 3(a), a minor positional deviation (e.g., 6 pixels diagonally) in the predicted bounding box of a small object results in a substantially significant greater reduction in IoU (from 1.00 to 0.29) compared to a similar deviation on a larger object (from 1.00 to 0.56). This observation underscores the heightened sensitivity of small objects to their predicted center coordinates. Concurrently, as demonstrated in Fig. 3(b), the IoU loss associated with HAR objects is more sensitive to the rotation angles of the objects. Therefore, it is potentially beneficial to adjust the weight during the training process to mitigate the disproportionate impact of rotational variations.

Recent work has sought to reduce localization deviations by modifying IoU-based losses. Approaches include incorporating geometric priors like RIoU [10], exploring angle-based regression losses [21], [22], [23], optimizing IoU-based techniques [24], [25], [26], [27], or reformulating angle prediction as a classification task. However, these methods often introduce complexity and treat center, size, and angle parameters as a strongly coupled set, thereby limiting fine-grained optimization control, particularly for small and HAR objects. While KFIoU [28] advances this direction by approximating the rotated box overlap with Gaussian-based intersections to avoid explicit angle parameterization, it still optimizes all bounding-box parameters (e.g., center point, size, angle) jointly, neglecting their distinct sensitivities across object types, and potentially leads to suboptimal convergence.

To address this parameter coupling, we propose dynamically weighted KFIoU (DKFIoU). DKFIoU embeds an area-aware and shape-aware, decoupled weighting strategy within the KFIoU framework, enabling independent adjustment of loss contributions for different parameters (e.g., center deviation and orientation) according to object characteristics. This decoupled and adaptive weighting yields more refined and interpretable regression, and provides a robust optimization paradigm that disentangles parameter sensitivities while preserving the geometric advantages of Gaussian-based intersections.

In summary, the contributions of this study are as follows.

1) We propose a multi-axis convolution network (MACNet) by integrating the LKDB and DC modules to synergistically enhance long-range contextual information and capture orientation-specific structural features, for robust small and HAR object detection in remote sensing imagery. This unified design differentiates our

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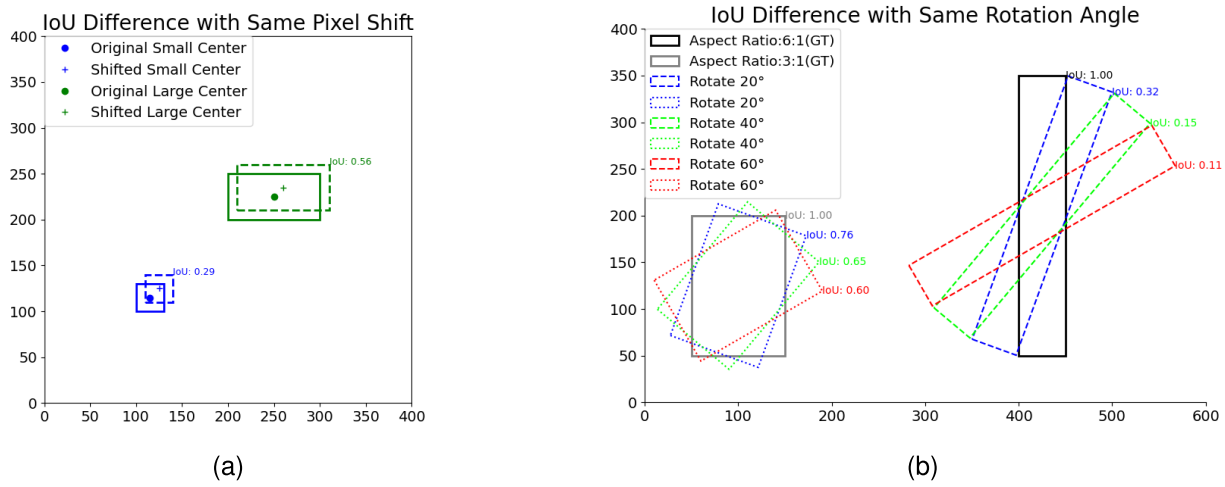


Fig. 3. (a) Variation in IoU resulting from pixel-level shift for objects of different sizes. The blue bounding box indicates small objects with dimensions less than 30×30 pixels, while the green bounding box represents larger objects exceeding 30×30 pixels. For small objects, minor pixel deviations can cause significant changes in IoU values. (b) Impact of rotation on IoU for objects with varying aspect-ratio, where the top-right represents a HAR object and the bottom-left denotes a low-aspect-ratio object. Specifically, the value of IoU for a HAR object decreased more rapidly (from 1 to 0.11), compared to the low-aspect-ratio object (from 1 to 0.60) after the same rotation angle. This indicates the IoU loss for an HAR object should be given greater weight in the training process.

MACNet from methods that improve only one aspect (context or shape modeling) and closes the gap in unified multichallenge mitigation for complex remote-sensing object detection.

2) To tackle scale- and orientation-sensitive localization dilemmas of small and HAR objects, we propose a DKFIoU loss. DKFIoU employs an adaptive weighting mechanism, correlated with object scale and aspect-ratio, to provide strengthened supervision for center-point regression in small objects and orientation-aware regression in HAR objects. This innovative design deviates from conventional fixed-global weighting paradigms, enabling more effective and interpretable bounding box optimization for remote sensing objects with diverse geometric characteristics.
3) Comprehensive experiment are conducted on DOTA-v1.0, DIOR-R, and HRSC2016 datasets. The results demonstrate that MACNet improves the accuracy of object detection. Specifically, our method achieves an improvement of 6.87% for small objects and 1.00% for HAR objects compared to the LSKNet baseline. These results highlight the effectiveness of our approach in addressing the specific challenges of RSOD. Ablation studies further validate the efficacy of our proposed modules in handling scale diversity, orientation variations, and complex spatial distributions inherent to geospatial object detection.

II. Related Work

A major challenge in RSOD is the accurate localization of small and HAR objects, as models struggle to extract highly distinguishable features from limited or distorted pixel information. Existing research tackling this challenge primarily focuses on three core directions: feature enhancement for small objects, optimized feature representation for HAR objects, and improved bounding box regression loss functions. Each direction has its limitations, which motivate the design of our proposed method.

A. Feature Enhancement for Small Object in Remote Sensing

Small object detection in remote sensing imagery is fundamentally challenging due to its limited pixel information, which impedes discriminative feature extraction and increases vulnerability to background clutter and noise. Most existing feature enhancement methods for small objects mainly fall into three categories: 1) multiscale feature fusion based on FPNs and their variants; 2) receptive-field expansion via large-kernel convolutions; and 3) attention-driven adaptive feature weighting. FPN-based approaches [13] and their variants, such as MB-FPN [29] and FPNFormer [12], leverage multiscale or bidirectional fusion to enhance feature information across different levels. However, the iterative top-down and bottom-up aggregation processes often dilute fine-grained details crucial for small objects.

Large kernel convolution models, including ConvNeXt [30] and LSKNet [15], capture richer contextual information by expanding the receptive field, thereby enhancing small object perception. Nevertheless, their fixed and position-invariant weights limit their adaptability, often leading to poor discrimination of small objects amidst cluttered backgrounds. To enhance adaptive feature weighting, attention mechanisms offer dynamic reweighting to emphasize the object-relevant features and suppress background noise [15], [16], [17]. However, many attention modules are integrated in a serial manner, which may cause feature attenuation and loss of delicate structural information. Consequently, a unified approach that simultaneously enhances contextual information and preserves fine-grained features for small objects remains necessary.

To address the limitations of existing methods, we design an LKDB module for parallel joint modeling of spatial and

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channel dependencies, enabling simultaneous preservation of fine-grained structure and enhancement of contextual information. This parallel design mitigates gradual information loss and feature attenuation prevalent in existing sequential methods, and yields a more robust representation for small objects in complex backgrounds.

B. Feature Representation Enhancement for HAR Objects in Remote Sensing

HAR objects are prevalent in remote sensing images; however, their accurate detection is severely hampered by standard square convolutional kernels. These kernels inherently lack directional adaptability, leading to the undesirable mixing of target features with background noise, particularly for HAR objects.

Existing methods primarily address this issue through deformable convolution, cascaded regression, and strip pooling strategies. Models like R3Det [7] employ deformable convolution to adapt to target geometric shapes; however, they often suffer from spatial offset field inconsistencies that disrupt directional feature integrity. DDDet [44] optimizes localization via multilevel cascaded regression, but its complex structure can lead to error accumulation and unstable performance for small and HAR objects due to angle-sensitive regression. The poly kernel network [14] enhances directional representation using strip convolutions and an inception mechanism, but its one-directional sequential extraction limits comprehensive multidirectional feature capture. Similarly, strip pooling-based methods [18], [19] efficiently encode directional features with $1 \times N$ and $N \times 1$ kernels but lack targeted orientation-aware adaptive optimization.

To overcome these limitations, we design a novel DC module. The DC module employs multidirection parallel strip pooling to explicitly encode object orientation information, thereby maintaining directional consistency and computational efficiency while significantly enhancing directional discriminative features. Furthermore, integrating this module within the shared backbone can simultaneously preserve small object features and suppress background noise. This produces a geometry-consistent and scale-robust feature representation that substantially improves HAR object detection without sacrificing small object performance, effectively circumventing the complexity and error accumulation issues inherent in existing deformable and cascaded approaches.

C. Bounding Box Regression Loss Optimization

Bounding box regression loss functions are pivotal for object detection model optimization. Traditional l_n-norm losses, including L1, L2, and Smooth L1 loss [22], [32], optimize the box parameters independently, ignoring the intrinsic correlation among center, size, and orientation, which leads to misaligned optimization objectives and evaluation metrics. Improved angle-sensitive variants of these losses similarly fail to resolve this fundamental issue.

IoU-based losses [33], [34], [35] optimize prediction overlap with ground truth, inherently considering parameter correlations, but face boundary discontinuity when calculating rotated box IoU near angle periodic limits, causing gradient instability [36]. Distribution-based losses such as KFIoU [28], GWD [37], and KLD [38] represent bounding boxes as Gaussian distributions to solve boundary discontinuity, yet KFIoU is highly sensitive to distribution scale, producing extremely weak optimization gradients for small objects with limited Gaussian variance, resulting in slow convergence and poor localization accuracy.

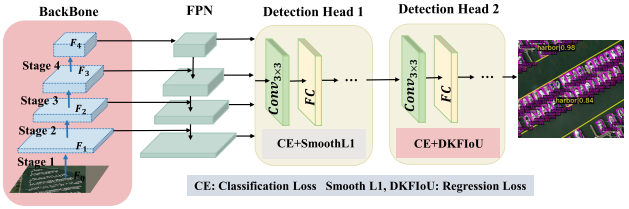


Fig. 4. Overall architecture diagram of the MACNet network includes the backbone, FPN, and head components.

To overcome this critical limitation, we propose the DKFIoU loss, which introduces an adaptive weighting mechanism to balance center-point regression and IoU overlap contributions based on target scale and aspect ratio. This dynamic adjustment enhances gradient response for small objects, stabilizes optimization for targets of varying sizes and geometries, and provides targeted supervision for both small and HAR objects, filling the optimization gap of existing distribution-based losses for small object detection.

III. Multi-Axis Convolution Network

To address feature attenuation in small objects and insufficient orientation awareness for HAR objects in remote sensing, we propose the MACNet, illustrated in Fig. 4. Shown in Fig. 5, the backbone is composed of four sequentially stacked stages, where each stage consists of two key submodules: the LKDB module, which uses large-kernel convolutions and dual-branch attention to mitigate feature loss for small objects, and the DC module, which employs strip convolutions to enhance orientation awareness for HAR objects. In addition, we introduce the DKFIoU loss, which provides a dynamic, geometry-aware supervision signal that adapts to object scale and orientation during training. This approach effectively guides the network to balance the challenge of resolving small objects with encoding accurate HAR orientations.

A. Large-Kernel Dual-Branch

To extract the discriminative features of remote sensing imagery, a LKDB module is proposed, which employs large-kernel convolutions to preserve as extensive a range of contextual information as possible. As illustrated in Fig. 5, the LKDB module initially performs multiscale large-kernel convolution to obtain the enriched contextual information. After that, the obtained feature maps X_n^1 and X_n^2 with different receptive fields are concatenated to obtain $\tilde{X}_n$, which are fed into dual branches, i.e., a spatial branch and channel branch. The information flow is formulated in the following equations:

$$X_n = BN\left(F_{n-1}\right) \tag{1}$$

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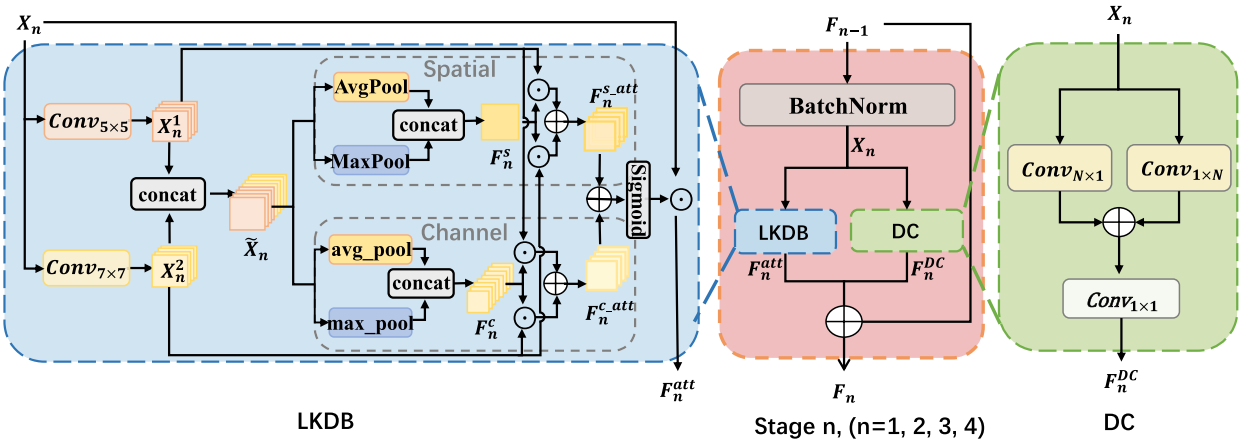


Fig. 5. Stage comprises two parts: the LKDB and DC. In the LKDB module, $Conv_{N\times N}$ denotes a large-kernel convolution with an N × N kernel. AvgPool and MaxPool refer to spatial average pooling and spatial max pooling that compress the channel dimension to 1. Similarly, avg_pool and max_pool represent channel average pooling and channel max pooling with a spatial dimension of 1 × 1. In the DC module, $Conv_{N\times1}$ and $Conv_{1\times N}$ represent $N \times 1$ and $1 \times N$ strip convolution, respectively. ⊕ denotes element-wise summation, ⊙ denotes element-wise multiplication.

$$X_n^1 = Conv_{1\times1}\left(Conv_{5\times5}\left(X_n\right)\right) \tag{2}$$

$$X_n^2 = Conv_{1\times1}\left(Conv_{7\times7}\left(X_n\right)\right) \tag{3}$$

$$\tilde{X}_n = CONCAT\left(X_n^1, X_n^2\right) \tag{4}$$

where $BN(\cdot)$ denotes batch normalization, $Conv_{K\times K}(\cdot)$ denotes convolution operations with kernel sizes of $K \times K$. $CONCAT(\cdot,\cdot)$ denotes a concatenation operation.

Motivated by the empirical finding that "exploiting both average and maximum pooling features greatly improves representation power of networks rather than using each independently" [31], an average pooling and a max pooling are simultaneously performed on $\tilde{X}_n$. In the spatial branch, the average and max pooling are performed along the channel, while in the channel branch, the average and max pooling are performed along the spatial dimension. To enhance the network's capability of focusing on the most relevant context information for object detection, both average and max pooling features are fused by concatenation and added in two branches to obtain a spatial attention matrix and a channel attention matrix, respectively.

Specifically, in the spatial branch, the spatial attention matrix, $F_n^{s_att}$, is as follows:

$$F_n^s = \sigma\left(Conv_{1\times1}\left(CONCAT \times \left(AvgPool\left(\tilde{X}_n\right), MaxPool\left(\tilde{X}_n\right)\right)\right)\right) \tag{5}$$

$$F_n^{s_att} = \sum_{i=1}^{2} X_n^i \odot F_n^s \tag{6}$$

where $\tilde{X}_n$ is a feature map with different receptive fields, which provide the multiscale context information. $AvgPool(\cdot)$ denotes channel-wise average pooling, which computes the mean of the C channel values at each spatial location (h, w) of the input feature map $\tilde{X}_n$. $MaxPool(\cdot)$ denotes channel-wise max pooling, which computes the maximum of the C channel values at each spatial location. Their dimension has changed from $C \times H \times W$ to $1 \times H \times W$. To dynamically guide the selection of appropriate receptive fields based on spatial priors in remote sensing images, we use the spatial attention feature mask F_n^s to reweight the feature maps X_n^1 and X_n^2 with different receptive fields. ⊙ denotes element-wise multiplication, $i = 1, 2$.

Similarly, to improve the network's ability to focus on discriminative channel-wise features for detecting objects, we employ a channel attention mechanism to enhance the feature maps produced by large kernel convolutions. The channel attention matrix, $F_n^{c_att}$, is obtained by the following equation:

$$F_n^c = \sigma\left(MLP\left(avg_pool\left(\tilde{X}_n\right)\right) + MLP\left(max_pool\left(\tilde{X}_n\right)\right)\right) \tag{7}$$

$$F_n^{c_att} = \sum_{i=1}^{2} X_n^i \odot F_n^c. \tag{8}$$

In order to distinguish from (5), which performs average pooling along channel, the $avg_pool(\cdot)$ is used to denote spatial-wise average pooling, which computes the mean across all $H \times W$ spatial positions for each channel. $max_pool(\cdot)$ denotes spatial-wise max pooling, which computes the maximum across all $H \times W$ spatial positions for each channel. Their dimension has changed from $C \times H \times W$ to $C \times 1 \times 1$. To obtain the attention weight for each channel, a multilayer perceptron (MLP) is employed to both avg-pooled and max-pooled descriptors. The MLP, which consists of two fully connected layers with a ReLU activation, is able to model nonlinear interchannel dependencies and to capture informative relationships across channels. A sigmoid function σ is used to generate a refined attention map F_n^c, which adaptively highlights the most relevant channel features while suppressing less informative ones. To dynamically guide the selection of appropriate receptive fields based on feature channels in remote sensing images, we use the channel attention feature mask F_n^c to adaptively reweight feature maps X_n^1 and X_n^2 along the channel dimension based on global context.

Intuitively, $F_n^{s_att}$ aggregates the two attention feature maps according to their spatial importance, while $F_n^{c_att}$ aggregates them according to their channel importance. To fully exploit both spatial and channel-wise discriminative features, we integrate the spatial attention feature map $F_n^{s_att}$ and channel

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attention feature map $F_n^{c_att}$ to produce the final feature map F_n^{att}, as shown in (9). This fusion strategy enables the network to simultaneously capture precise localizations and semantic relevance, benefiting from the complementary strengths of the dual-branch architecture

$$F_n^{att} = X_n \odot \sigma\left(F_n^{s_att} \oplus F_n^{c_att}\right) \tag{9}$$

where F_n^{att} is the feature maps that are weighted by context information with different receptive fields by CNN-attention mechanism [31].

B. Direction-Aware Convolution

To explicitly perceive and utilize the directional information of HAR objects, the DC module is introduced by focusing on orientation-specific feature extraction. The DC module employs two parallel strip convolutions (e.g., $1 \times N$ and $N \times 1$) to encode directional cues along dominant axes, thereby reinforcing the network's ability to represent HAR objects and maintain geometric integrity during localization.

The formal definition of the proposed DC module is given in the following equation:

$$F_n^{DC} = Conv_{N\times 1}(X_n) \oplus Conv_{1\times N}(X_n). \tag{10}$$

The output feature of nth stage module is obtained by aggregating F_{n-1}, F_n^{att}, and F_n^{DC}

$$F_n = F_{n-1} \oplus F_n^{att} \oplus F_n^{DC} \tag{11}$$

where F_{n-1} denotes a residual term, F_n^{att} incorporates the multiscale context information, and F_n^{DC} collects long-range dependencies and local details; $\oplus$ denotes element-wise summation.

We integrate cross-directional strip convolutions within the DC module, which operates in parallel with the LKDB module to complement global–local feature enhancement. This parallel design enables the network to jointly capture multi-axis contextual dependencies and complex structural patterns in remote sensing images, thereby improving feature representation for both small and HAR objects. The two modules (LKDB and DC) are stacked in parallel and repeated four times to construct the backbone, allowing early stage modeling of global context while maintaining robust discriminative features across diverse object scales and shapes.

C. DKFIoU Loss

While the LKDB and DC modules enhance the network's ability to extract discriminative and direction-aware features, the final detection accuracy also depends on how effectively the regression loss guides localization. However, existing IoU-based losses (e.g., SkewIoU or KFIoU) usually exhibit unstable optimization for small or elongated objects due to their limited ability to balance scale-sensitive and angle-sensitive gradients. To overcome this limitation, we further introduce a DKFIoU loss that adaptively adjusts optimization focus according to object characteristics.

The KFIoU loss function is an easy-to-implement approximation of the SkewIoU loss, which is widely used in rotation detection [39]. It estimates the overlap between predicted and ground-truth bounding boxes through a Gaussian product formulation. Specifically, after generating the predicted bounding box parameters, the ground-truth bounding box and predicted bounding box are defined by $GT(x_{gt}, y_{gt}, w_{gt}, h_{gt}, \theta_{gt})$ and $P(x_p, y_p, w_p, h_p, \theta_p)$, where (x, y) are the bounding-box center coordinates, (w, h) its width and height, and θ its rotation angle.

Since the value of (x, y, w, h) are related to the specific, while angles are measured in radians within the range $[-\pi/2, \pi/2]$, their physical interpretations and data ranges of the angle θ, position (x, y), and size (w, h) differ. KFIoU unified the range of all parameters of an oriented bounding box by normalizing them using Gaussian distributions $\mathcal{G}(\mu, \Sigma)$, which is defined by the following equation:

$$\Sigma = R \wedge R^{\top}, \quad \mu = (x, y) \tag{12}$$

where (x, y) denote the coordinates of the bounding box center. R represents the rotation matrix, and $\wedge$ represents the diagonal matrix of eigenvalues, which are defined by the following equation:

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \wedge = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix} \tag{13}$$

where w and h denote the width and height of the bounding box, θ denotes the angle of the bounding box.

After transforming a bounding box into a Gaussian distribution, we can employ the KFIoU to calculate the overlapping area between the predicted and ground-truth bounding boxes. Motivated by [28], the product of two Gaussian distributions is used to denote the distribution of the overlapping based on the following equation:

$$s_g G_{kf}(\mu_{ov}, \Sigma_{ov}) = \mathcal{G}_{gt}(\mu_{gt}, \Sigma_{gt})\, \mathcal{G}_p(\mu_p, \Sigma_p) \tag{14}$$

where $\mathcal{G}_{kf}(\mu_{ov}, \Sigma_{ov})$ denotes a new Gaussian distribution, its mean and variance are calculated by the following equation:

$$\begin{aligned} \mu_{ov} &= \mu_{gt} + K(\mu_p - \mu_{gt}) \\ \Sigma_{ov} &= \Sigma_{gt} - K\Sigma_{gt} \\ K &= \Sigma_{gt}\left(\Sigma_{gt} + \Sigma_p\right)^{-1} \end{aligned} \tag{15}$$

where K denotes a Kalman gain and s_g is a scale factor, which is calculated by the following equation:

$$s_g = \frac{1}{\sqrt{2\pi\left(\Sigma_{\text{gt}}^2 + \Sigma_p^2\right)}} \exp\left(-\frac{\left(\mu_{\text{gt}} - \mu_p\right)^2}{2\left(\Sigma_{\text{gt}}^2 + \Sigma_p^2\right)}\right). \tag{16}$$

As shown in (15), s_g is related to the Euclidean distance between the centers of the two Gaussian distributions $(\mu_{gt} - \mu_p)$. Since s affects the normalization of the resulting distribution, we first introduce a center-point constraint loss, Smooth L_1, to explicitly minimize the distance between the two centers during training. As shown in Fig. 6, this is achieved by the first detection head at the initial stage, effectively aligning the centers so that s_g can be approximated as a constant in subsequent computations. Under this assumption, the area of Gaussian distributions can be approximately calculated by $Area(\Sigma) = 4|\Sigma|^{1/2}$ [25]. This design ensures that the loss function remains optimizable even in nonoverlapping scenarios.

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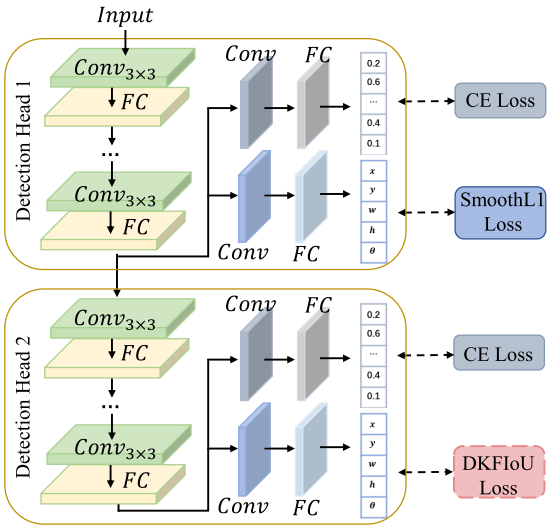


Fig. 6. Structural diagram of the detection head. CE loss stands for cross-entropy loss, used to supervise classification tasks.

Therefore, the IoU of the predicted and ground-truth bounding boxes is calculated by the following equation:

$$KFIoU = \frac{Area\left(\Sigma_{ov}\right)}{Area\left(\Sigma_{gt}\right) + Area\left(\Sigma_{p}\right) - Area\left(\Sigma_{ov}\right)} \tag{17}$$

where Σ_{ov} represents the variance of the Gaussian distribution of overlapping between the predicted and ground-truth bounding boxes, Σ_{gt} represents the variance of the ground-truth bounding box, and Σ_{p} represents the variance of the predicted bounding box.

Then, the loss function of KFIoU is represented by the following equation:

$$L_{kf}\left(\Sigma_{gt}, \Sigma_{p}\right) = e^{1-KFIoU} - 1. \tag{18}$$

To obtain high-performance detection in the scenario containing various types of objects, an effective regression loss should be defined to modulate the contribution of each parameter based on the types of objects to be detected. As an example, the center point parameters (x, y) are important for small object detection as they directly influence the localization accuracy of these objects within the image. In contrast, the parameters of angle, width, and height (θ, w, h) play an important role in HAR object detection since misalignment in these features might lead to significant IoU degradation. Therefore, a DKFIoU loss function was designed to encourage the learning process to be self-modulated by adaptively adjusting the contribution of each parameter.

Specifically, an area-aware weight A_j is incorporated into the regression loss to modulate the contribution of objects with varying scales (width w and height h). To ensure smoother optimization, it is defined by the square root of the object area, based on the following equation:

$$A_j = \frac{1}{\sqrt{w_j h_j + \epsilon}} \tag{19}$$

where A_j represents the weight of the jth object and ϵ is a small positive real number.

Simultaneously, to incentivize the model to learn more robust representations, a shape-aware weight, R_j, is defined in (20) based on the aspect ratio of the object. It adjusts the contribution of the IoU loss of objects with HARs in the regression loss

$$R_j = \frac{r_j}{1 + r_j} \tag{20}$$

where R_j represents the aspect ratio weight of the jth object and r denotes the aspect ratio of the object, which is calculated by $r_j = (\max(w_j, h_j))/(\min(w_j, h_j))$.

Therefore, the DKFIoU loss is defined by the following equation:

$$\begin{aligned} DKFIoUL_r\left(\mathcal{G}\left(b_j\right), \mathcal{G}\left(gt_j\right)\right) &= R_j L_{kf}\left(\Sigma_{gt}, \Sigma_{p}\right) \\ &\quad + A_j L_c\left(\mu_{gt}, \mu_{p}\right) \end{aligned} \tag{21}$$

where L_{kf} represents the KFIoU between the ground truth and the predicted bounding box, and L_c is the distance between the ground truth and the predicted bounding box, which is calculated using Smooth L1 loss.

Therefore, the overall loss of MACNet is formulated in the following equation:

$$\begin{aligned} L = &\frac{1}{N_{\text{pos}}} \sum_{j=1}^{N_{\text{pos}}} DKFIoUL_r\left(\mathcal{G}\left(b_j\right), \mathcal{G}\left(gt_j\right)\right) \\ &+ \frac{1}{N} \sum_{j=1}^{N} L_{\text{cls}}\left(p_j, t_j\right) \end{aligned} \tag{22}$$

where N_{pos} and N indicate the number of positive anchors and that of all anchors. b_j and gt_j denote the jth predicted and ground-truth bounding box. $\mathcal{G}(\cdot)$ is a Gaussian transfer function. $L_{cls}(p_j, t_j)$ represents the classification loss, t_j and p_j represent the ground-truth label and the predicted label of the jth object. The classification loss L_c is measured by a cross-entropy loss.

IV. Experiments

A. Experiments Setting

1) Dataset:

a) DOTA-v1.0 [40]: The DOTA v1.0 dataset is one of the most widely used benchmark datasets in the field of remote sensing image object detection, renowned for its large scale and diversity. This dataset comprises 2806 aerial images with resolutions ranging from 800 × 800 to 4000 × 4000 pixels and contains a total of 188 282 annotated target instances across 15 object categories. These categories encompass a wide range of targets such as vehicles, ships, and buildings, as well as elongated targets like bridges. The dataset provides refined annotations with oriented bounding boxes, supporting the evaluation of algorithms capable of addressing challenges such as orientation variation, small objects detection, and dense target distribution. DOTA v1.0 captures complex real-world scenarios and serves as a vital resource for advancing the technology of high-resolution remote sensing image object detection.

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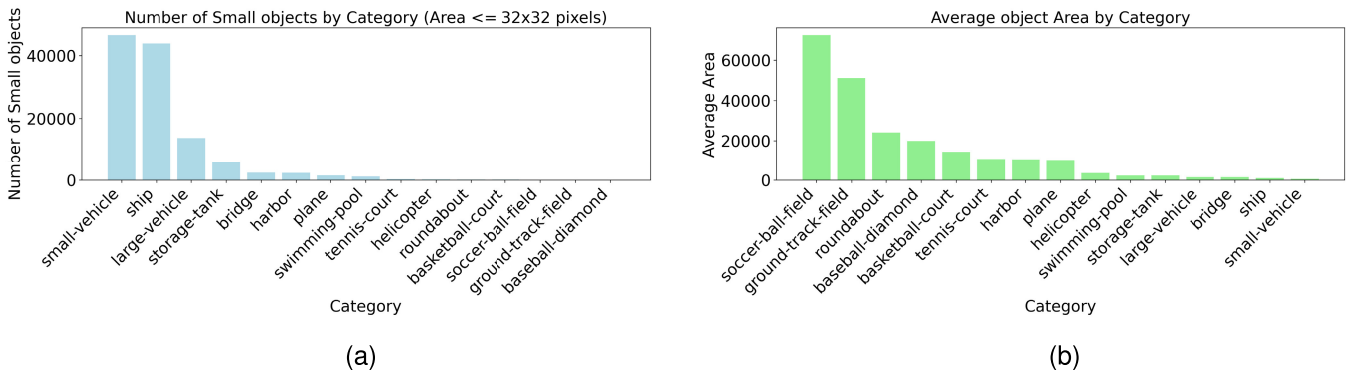


Fig. 7. In the DOTA v1.0 dataset, categories with smaller average object sizes or larger counts of small instances were selected for comparison to better evaluate the model's performance on small-scale objects. (a) Number of Small objects by each category. (b) Average area for each category.

b) DIOR-R [41]: The DIOR-R dataset is a comprehensive and challenging benchmark specifically designed for remote sensing image object detection. It comprises 23 463 high-resolution aerial images with dimensions ranging from 800 × 800 to 1000 × 1000 pixels, covering a diverse collection of 20 object categories. These categories include both small and large targets such as vehicles, airplanes, and tanks, as well as elongated targets like bridges and ships. The targets within the dataset exhibit a variety of orientations, sizes, and complex backgrounds, providing a true reflection of real-world scenarios. The precise annotations of oriented bounding boxes support the evaluation of detection methods capable of addressing issues such as localization accuracy, aspect ratio variation, and orientation sensitivity. DIOR-R serves as a critical resource for advancing detection algorithms in diverse and complex remote sensing image scenes.

c) HRSC2016 [42]: The HRSC2016 dataset is a benchmark dataset widely used in the field of object detection, particularly for ship detection in high-resolution remote sensing images. It contains 1061 aerial images with a resolution ranging from 300 × 300 to 1500 × 900 pixels. These images were collected from various locations, capturing ships in different orientations, scales, and aspect ratios, which present significant challenges for detection algorithms. The dataset provides precise annotations, including rotated bounding boxes and the corresponding angle information, enabling the development and evaluation of methods targeting HAR objects and varying orientations. HRSC2016 is considered a standard dataset for validating the effectiveness of detection algorithms in handling complex object geometries and challenging environmental conditions.

2) Implementation Details: To evaluate the effectiveness of the proposed method, we conduct comparison experiments with different SOTA baselines on DOTA-v1.0, DIOR-R, and HRSC2016 datasets, respectively. The baselines include one-stage and two-stage architectures. In addition, we evaluate recent proposed transformer-based methods such as DETR and ARS-DETR, which represent the latest progress in query-based detection paradigms.

To ensure fairness, we adhered to the same dataset processing methods as other mainstream approaches. For the DOTA v1.0 and DIOR-R datasets, we cropped the images into subimages of 1024 × 1024 pixels. On the HRSC2016 dataset, we rescaled the images by setting the longer side to 800 pixels without altering the aspect ratio. The proposed MACNet backbones are pretrained on ImageNet for 300 epochs on all three datasets. We utilized the AdamW optimizer to train the model for 36 epochs on the HRSC2016 dataset and for 12 epochs on the DOTA v1.0 and DIOR-R datasets. The model was implemented under the MMRotate [43] framework, which is specifically designed for oriented object detection tasks and is capable of effectively handling directional targets. We conducted model training and testing using a single RTX3090 GPU.

To objectively validate the effectiveness of the proposed method on small-scale and HAR objects, we conducted a statistical analysis of the dataset. Specifically, we calculated the average object size of each category and ranked them from large to small, as well as the number of small objects (defined as those with an occupied area $\leq 32 \times 32$ pixels) in each category. As shown in Fig. 7, in the DOTA v1.0 dataset, categories such as small vehicle and ship exhibit significantly smaller average object sizes or contain a large number of small-scale instances. Therefore, these categories were selected for further comparative analysis to better reflect the model's capability in small object detection.

B. Main Results

1) Results on DOTAv1.0: We conducted comparison experiments with 19 widely used OD approaches to evaluate the performance of our proposed model on the DOTA v1.0 dataset. The experimental results in Table I show that our proposed model outperforms the other methods in terms of mean average precision (mAP). Table I shows that our model achieved a 4.56% improvement over Roi-Trans (baseline) in terms of mAP. Compared to LSKNet, which is an attention-based detection model, our method performed significantly better than LSKNet on small object and HAR objects detection, with a mAP improvement of 6.87% and 1.00% on small object detection and HAR objects detection, respectively.

Compared to DDDet with a complex structure, our proposed model obtains slightly decreased results on DOTA v1.0. However, it is worth noting that our approach obtains the highest

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TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET. C1: PLANE, C2: BASEBALL-DIAMOND, C3: BRIDGE, C4: GROUND-TRACK-FIELD, C5: SMALL-VEHICLE, C6: LARGE-VEHICLE, C7: SHIP, C8: TENNIS-COURT, C9: BASKETBALL-COURT, C10: STORAGE-TANK, C11: SOCCER-BALL-FIELD, C12: ROUNDABOUT, C13: HARBOR, C14: SWIMMING-POOL, AND C15: HELICOPTER. NOTES: "ROTATED FASTER R-CNN" IS ABBREVIATED AS "RFRCNN" AND "DETR-O-CSL" IS ABBREVIATED AS "D-O-CSL"

method	mAP(%)	class mAP(%)														
		c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15
R3Det [7]	71.17	88.96	76.99	47.09	70.89	77.54	76.19	86.24	90.91	79.45	83.6	52.98	62.5	64.65	67.32	42.29
GWD [37]	69.25	88.68	78.59	45.41	71.46	72.27	68.26	77.05	90.8	80.56	81.93	46.48	60.14	63.87	67.39	46.06
KLD [38]	69.64	88.27	76.22	46.22	72.73	72.11	67.84	77.63	90.77	80.67	83.03	52.74	62.23	64.91	65.95	43.22
CFA [21]	69.63	88.07	74.57	49.25	74.43	79.02	74.14	86.76	90.87	80.39	86.03	48.49	58.89	64.38	66.87	22.51
SASM [45]	66.46	80.80	66.37	45.41	65.58	72.08	72.35	79.83	90.90	75.69	83.06	49.06	59.47	63.76	59.42	33.19
CSL [53]	69.49	89.34	79.68	40.86	69.94	77.69	62.04	77.49	90.87	82.88	82.07	59.97	65.30	53.56	64.04	46.61
S2ANet [21]	74.12	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94
ReDet [46]	76.25	88.79	82.64	53.97	74.00	78.13	**84.06**	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59
R_Trans [47]	73.12	88.34	77.07	51.63	69.62	77.45	77.15	87.11	90.75	84.90	83.14	52.36	66.66	66.31	69.99	54.40
RFRCNN [48]	69.05	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18
ORCNN [49]	75.87	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28
SCRDet [50]	72.61	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	86.86	65.02	66.68	66.25	68.24	65.21
DETR-O [57]	69.48	84.89	70.71	46.04	61.92	73.99	78.83	87.71	90.07	77.97	78.41	47.07	54.48	66.87	67.66	55.62
D-O-CSL [53]	72.15	86.27	76.66	46.64	65.29	76.80	76.32	87.74	90.77	79.38	82.36	54.00	61.47	66.05	70.46	61.97
ARS-D [51]	74.16	86.97	75.56	48.32	69.20	77.92	77.94	87.69	90.50	77.31	82.86	60.28	64.58	74.88	71.76	66.62
KFIoU [28]	77.35	89.46	85.72	54.94	80.37	77.16	69.23	80.90	90.79	87.79	86.13	**73.32**	68.11	75.23	71.61	**69.49**
DFDet [22]	74.71	88.92	79.25	48.40	70.00	80.22	78.85	87.21	90.90	83.13	83.98	60.07	66.49	68.27	76.78	58.11
LSKNet [15]	77.49	89.66	85.52	57.72	75.7	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82
PKINet [14]	77.17	89.33	85.59	55.75	74.69	74.69	79.13	88.05	90.90	87.43	86.90	61.67	64.25	77.77	75.38	66.08
DDDet [44]	**79.41**	89.55	**85.88**	**57.93**	**80.38**	79.56	83.16	89.13	90.85	86.94	**87.40**	68.51	**71.30**	**78.41**	**79.29**	62.84
MACNet(our)	77.68	**90.64**	84.75	57.80	74.96	**81.82**	78.77	**89.24**	**91.10**	86.98	85.22	60.93	66.59	76.98	76.22	63.28

TABLE II

COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET. NOTES: "ROTATED FASTER R-CNN" IS ABBREVIATED AS "RFRCNN" AND "GLIDING VERTEX" IS ABBREVIATED AS "G.V"

method	mAP(%)	Vehicle	Ship	Windmill	Bridge	Harbor
RetinaNet [32]	55.06	38.30	75.34	63.23	18.86	32.42
R_Trans [47]	63.87	43.27	81.21	65.52	40.68	**47.71**
RFRCNN [48]	59.54	50.46	81.14	65.21	34.20	34.95
DCFL [56]	66.80	50.9	88.61	**70.90**	42.10	39.25
G.V [52]	60.06	47.71	80.22	65.04	33.88	37.22
FCOS-O [58]	59.80	42.07	80.44	64.95	34.81	34.10
DFDet [22]	62.11	44.11	80.40	63.35	34.11	38.26
ARS-D [51]	66.12	50.57	84.55	70.28	41.13	36.33
LSKNet [15]	65.90	65.24	88.41	65.68	41.56	42.53
PKINet [14]	67.30	66.31	86.92	64.21	42.76	46.82
MACNet(ours)	**68.47**	**70.40**	**90.00**	64.50	**44.10**	46.70

TABLE III

COMPARISON RESULTS ON THE HRSC2016 DATASET

Method	Roi_Trans [47]	G.V [52]	R3det [7]	CSL [53]	GWD [37]
mAP(%)	86.2	88.2	89.26	89.62	89.85
Method	KLD [38]	LSKNet [15]	S2ANet [21]	Oriented Rep [55]	ORCNN [49]
mAP(%)	89.97	90.1	90.17	90.3	90.4
Method	ReDet [46]	PKINet [14]	DDDet [44]	MACNet(ours)	
mAP(%)	90.46	90.65	90.28	**90.7**	

AP for typical small objects (e.g., small-vehicle and ship) and HAR objects (e.g., bridge). These results validate the design purpose of MACNet: the LKDB module preserves extensive contextual information and key details of small objects, the DC module enhances the directional perception for HAR objects, and the DKFIoU loss adaptively balances scale-sensitive and shape-sensitive gradients during regression. Furthermore, the DC module based on stripe convolution provides a direction-aware representation that better preserves the directionality of HAR objects, further eliminates the noise interference introduced by square convolution, and enhances the features of small objects.

2) Results on DIOR-R: To further verify the performance of the proposed model on a more complex and diverse dataset, we conduct experiments on the DIOR-R dataset, which provides a more realistic evaluation environment compared to a simpler and controlled dataset.

The comparison results in Table II demonstrate that the proposed MACNet achieves the best performance with a mAP of 68.47%, closely followed by PKINet, which achieves a mAP of 67.30%. In specific categories, MACNet ranks first in the vehicle category with a mAP of 70.40%, and also leads in the ship category with a mAP of 90.0%, and achieved a performance of 44.10% in the bridge category. These findings indicate that our method performs well on diverse and complex datasets and shows performance improvements in small object detection and HAR object detection. This is attributed to the fact that MACNet can extract effective discriminative object features through LKDB and DC. However, for the Windmill category, DCFL performs best with a mAP of 70.90%. These results suggest the dynamic coarse-to-fine label assignment of DCFL enhances the detection of variably oriented windmills. Overall, the MACNet has shown significant competitiveness in multicategory object detection tasks on the DIOR-R dataset, particularly excelling in the Vehicle, Ship, and Bridge categories.

3) Results on HRSC2016: To further evaluate the capability of our method on HAR targets, we conduct experiments on the HRSC2016 benchmark; results are summarized in Table III. The results show that MACNet outperforms other models, including the DDDet and R3det, achieving a mAP of 90.7%.

These results demonstrate that MACNet is robust to the extreme aspect ratios and dense object layouts characteristic of HRSC2016. We attribute the improvement to the

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TABLE IV
ABLATION RESULTS ON INFLUENCE OF EACH INDIVIDUAL MODULES

	R50	MACNet			Class mAP(%)				mAP(%)
		LKDB		DC	small objects		HAR objects		
		Spatial	Channel		Small-Vehicle	ship	bridge	harbor	
1	✓				77.45	87.11	51.63	66.31	73.12
2		✓			80.14	88.01	53.17	69.92	74.77
3			✓		79.88	88.21	54.78	72.41	75.13
4				✓	78.61	88.24	54.11	71.32	74.41
5		✓	✓		80.21	87.96	54.24	71.86	74.82
6		✓		✓	80.54	88.48	55.12	72.13	75.60
7			✓	✓	80.62	88.29	55.37	73.50	75.33
8		✓	✓	✓	81.20	88.80	55.94	74.88	76.47

complementary design of the model, which improves localization accuracy for both small and HAR objects.

C. Ablation Study

To evaluate the effectiveness of each component of our proposed method, we conduct a series of ablation experiments. These experiments are designed to quantify the contributions of different modules and settings in the model.

To investigate the contribution of each individual module, we systematically remove or replace each component of the proposed model. The ablation results are illustrated in Table IV. In summary, Table IV shows that the *LKDB* module plays a more important role in improving detection accuracy, compared to the *CD* module. And within the *LKDB* module, the *Spatial* attention is more important than *Channel* attention for identifying the small objects, while the *Channel* attention is more critical for identifying the HAR objects. The detailed analysis is listed below.

1) Baseline: The baseline uses ResNet50 as the backbone network with FPN for multiscale feature extraction, as shown in row 1 of Table IV.

a) Spatial Attention Module Only (Spatial): Adding the spatial attention module to the baseline improves the mAP by 1.65%, as shown in row 2 of Table IV. It indicates that focusing on spatial features helps the model to distinguish small objects from complex backgrounds in remote sensing.

b) Channel Attention Module Only (Channel): Integrating the *Channel* Attention module results in a slight performance increase over using only the *Spatial* Attention module, improving the mAP by 2.01% relative to the baseline, as presented in row 3 of Table IV. This indicates that incorporating the *Channel* Attention Module can enhance the model's feature representation ability by highlighting significant channels and reducing background noise. In addition, we observe that the *Channel* module yields improvements of 3.15% and 6.10% in mAP for bridge and harbor detection, respectively. It proves particularly effective for detecting objects with HARs, as it can emphasize relevant feature channels—such as edges or orientations—associated with these objects, thereby leading to better detection accuracy.

c) DC Module Only (DC): The addition of *DC* results in a modest increase in performance with a 1.29% average improvement in mAP, as shown in row 4 of Table IV. Compared to the baseline, the *DC* module contributes to 5.01% improvement on bridge (HAR object) detection, while improving the mAP by 1.16% on small-vehicle detection. This finding indicates that capturing directional dependencies contributes positively to the HAR objects detection accuracy by enhancing the ability of modeling structural information of different objects.

d) LKDB Module (LKDB): Combining dual-branch with large kernel provides a cumulative benefit over using one alone, achieving in mAP by 1.70% compared to baseline, as shown in row 5 of Table IV. This result is slightly higher than that of the *Spatial* attention module alone (74.77%), with an improvement of 0.05%, especially for the HAR objects (e.g., achieved an improvement of 1.07% in mAP on bridge detection). For small objects (e.g., small-vehicle), employing the dual-branch improves the mAP from 79.88% (*Channel* only) and 80.14% (*Spatial* only) to 80.21%, respectively. It is interesting that the performance based on the dual-branch is lower than the one based on the *Channel* attention module only as well. These findings indicate that while the *LKDB* module provides marginal benefits over spatial attention alone, it does not fully capitalize on the advantages of *Channel* attention. This suggests potential redundancy or suboptimal interaction between the two modules, highlighting the need for more refined fusion strategies to unlock their full complementary potential, which further validates the design choice of the *DC* module, as a more refined integration could unlock greater performance gains.

e) DC and Spatial: As shown in row 6 of Table IV, this combination outperforms the standalone *DC* module by 1.19%. For small objects, Small-Vehicle and Ship detection accuracy are increased by 1.93% and 0.27%, respectively, indicating that *Spatial* attention effectively enhances the localization of salient regions and reinforces spatial context information of small objects. Meanwhile, it surpasses the single *Spatial* attention module by 0.83%, and improves 1.95% and 2.21% in detecting HAR objects, demonstrating that the *DC* module provides valuable orientation-sensitive structural information. In addition, this design achieves a 2.48% mAP increase compared to the baseline, suggesting that the integration of *DC* and *Spatial* attention enables the model to capture spatial distribution information. This joint design enhances the representation of fine-grained and arbitrary orientations of objects in remote sensing imagery.

f) DC and Channel: As shown in row 7 of Table IV, this design achieves a 0.92% increase in mAP compared to the *DC* only module, indicating that the introduction of channel attention effectively enhances feature weighting and helps suppress background noise. Meanwhile, it outperforms the standalone *Channel* attention module by 0.20%. For the HAR objects, bridge and harbor are increased by 0.59% and 1.09%, respectively, suggesting that incorporating *DC* introduces critical orientation cues that *Channel* attention alone cannot capture. In addition, this combination achieves a mAP improvement of 2.21% over the baseline; this result highlights the complementary nature of *Channel* and *DC* in enhancing detection performance.

g) Complete MACNet: Utilizing the complete *MACNet* backbone significantly outperforms all previous configurations, attaining an mAP of 76.47%, as shown in row 8 of

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TABLE V

ABLATION STUDY ON THE STRUCTURE OF THE DUAL-BRANCH. " SPATIAL → CHANNEL " INDICATES A SEQUENTIAL ATTENTION OPERATION, WITH SPATIAL ATTENTION PRECEDING CHANNEL ATTENTION. " CHANNEL → SPATIAL " INDICATES A SERIAL ATTENTION OPERATION, WITH CHANNEL ATTENTION PRECEDING SPATIAL ATTENTION. " SPATIAL AND CHANNEL " REPRESENT PARALLEL ATTENTION OPERATIONS

	Spatial → Channel	*Channel → Spatial*	*Spatial and Channel*
mAP(%)	77.28	77.26	77.58

TABLE VI

ABLATION STUDY OF LOSS FUNCTION ON DOTA V1.0 DATASET

Loss function	Base Loss	$+A_j$	$+R_j$	Small Object		HAR Object		mAP(%)
				Small-Vehicle	ship	bridge	harbor	
$l_n - norm$	$l_n - norm$			81.20	88.8	55.94	74.88	76.47
KFIoU	KFIoU			80.90	88.7	56.35	75.23	76.94
DKFIoU	KFIoU	✓		81.74	89.33	56.64	76.37	77.46
DKFIoU	KFIoU	✓	✓	81.82	89.24	57.8	76.98	77.58

Table IV. Compared to the baseline, the complete *MACNet* achieves a 3.35% increase in mAP, which indicates that the joint design of directional, spatial, and channel attention mechanisms enables more comprehensive feature representation and therefore enhances the detection accuracy of small objects and HAR objects in remote sensing.

Ablation Study on Dual Branch:

To further examine the impact of the dual-branch's architecture on detection performance, we developed both sequential and parallel structures to facilitate the implementation of channel and spatial attention, respectively. The results are shown in Table V.

Experimental results reveal that the integration strategy of the attention mechanism within LKDB influences the performance of the model. Specifically, as shown in Table V, within the sequential structure, *Spatial* → *Channel* achieved 77.28% mAP, while *Channel* → *Spatial* decreased the detection accuracy to 77.26%. This different performance implies that preprocessing through channel attention might inadvertently impede the subsequent effective refinement of spatial contextual features. In contrast, the concurrent operation of parallel attention mechanisms leverages the inherent synergy between spatial and channel attention to simultaneously optimize both spatial context information and feature discriminability, resulting in enhanced performance.

Ablation Study on DKFIoU:

To verify the efficacy of the proposed loss function design, we conducted ablation studies on the DOTA-v1.0 dataset, with the results summarized in Table VI.

Table VI presents the comparative results of different regression losses. Introducing area-aware weighting A_j into the KFIoU formulation yields an mAP improvement over the l_n − *norm* loss and the standard KFIoU. In particular, the accuracy for small objects such as Small-Vehicle and Ship increases by 0.84% and 0.63%, respectively. These results confirm that incorporating object-scale information provides more informative supervision, which effectively stabilizes the regression of small objects and enhances their localization precision. By jointly considering the scale and shape of objects, the proposed DKFIoU achieves the highest overall accuracy, with a 1.11% mAP gain compared to the l_n − *norm* loss and a 0.64% improvement over the standard KFIoU. Compared with its ablated variant without shape-aware weighting (i.e., DKFIoU w/o R_j), DKFIoU further improves mAP. Notably, for elongated categories such as Bridge, the detection performance rises by 1.16%. Importantly, the inclusion of shape-aware weighting does not degrade small object detection, which indicates that DKFIoU maintains robust performance across both small objects and HAR objects.

These consistent improvements demonstrate that the dynamic weighting mechanism in DKFIoU not only adapts to object scale and shape to enhance overall detection accuracy but also mitigates the weak-gradient and distribution discrepancy issues that limit the original KFIoU by providing stronger supervision for small objects and HAR objects, respectively.

The qualitative analysis of the performance of MACNet is provided, compared to the SOTA baseline (LSKNet) on the DOTA dataset. The visualized results are illustrated in Fig. 8.

Fig. 8 illustrates the qualitative comparisons across three representative scenes, e.g., parking, architecture, and road. We compare three SOTA OD methods, that is, LSKNet, PKINet, and the proposed MACNet. It can be observed that three methods have excellent performance in detecting large objects (e.g., the tennis court in the third line). However, our method shows superior performance on the detection of small and HAR objects. For instance, in the parking scene, our proposed method can successfully identify the objects missed by LSKNet and PKINet (as shown in red boxes) and false detections by PKINet (shown by orange boxes). Similar results are shown in the architecture and road scenes. However, there is still room for improvement in detecting small objects partially obscured by shadows, such as vehicles in the second row and the third row obstructed by trees (the small vehicle in the red box). The performance gain is attributable to the LKDB integration within MACNet: the large kernel enlarges the receptive field to capture broader contextual cues; the dual-branch design suppresses irrelevant background while amplifying object feature responses, particularly for small objects.

Similarly, Fig. 9 presents representative visualization results of HAR objects detection on the DOTA dataset. Fig. 9 presents qualitative results for three representative HAR scenes (harbor, grayscale, and ship). Compared with our method, LSKNet and PKINet exhibit more false detections (orange bounding boxes) and missed detections (red bounding boxes). For example, PKINet fails to detect the harbor in the first row, and LSKNet incorrectly splits the bridge into two bounding boxes in the second row. While MACNet generates continuous detection results with fewer misses, its directional localization of the harbor (first row) still has room for improvement. This improved performance can be attributed to the DC module in MACNet, which is particularly beneficial for detecting HAR objects by efficiently capturing directional dependencies. This finding has been verified by the previous ablation study results, showing a significant drop in performance on HAR objects when the DC module is removed.

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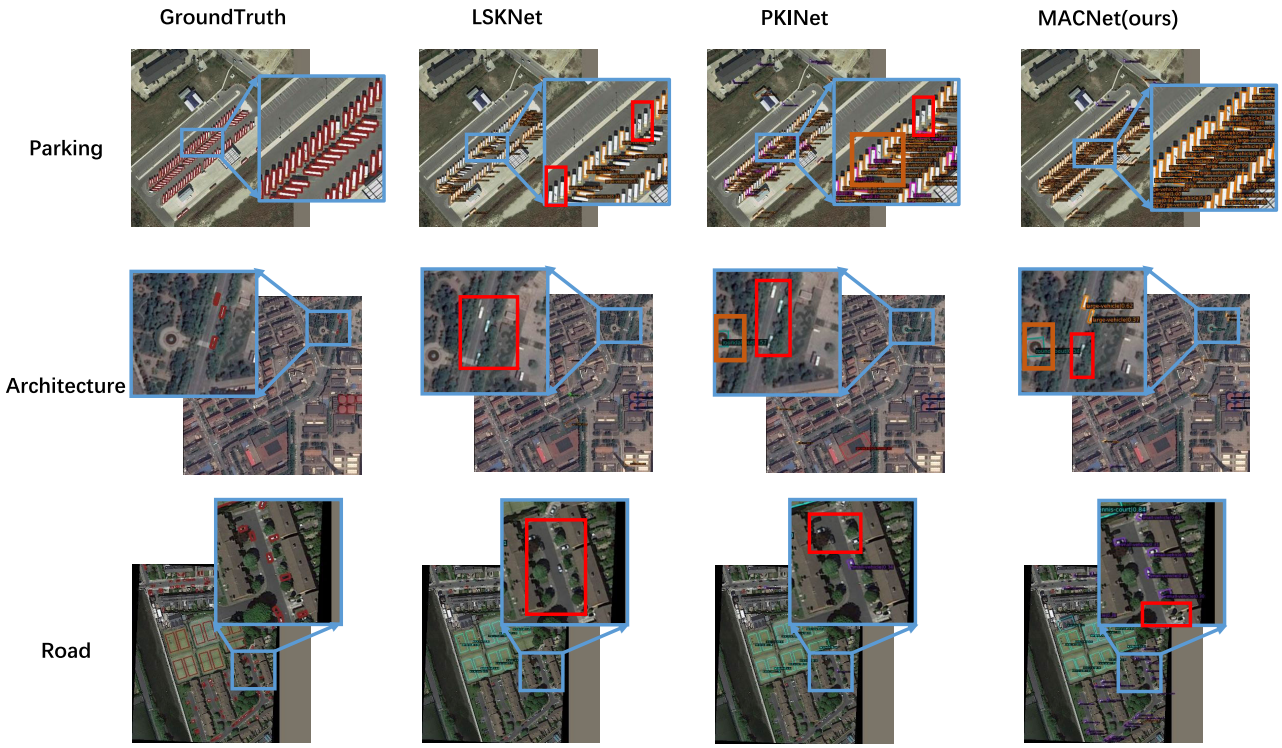


Fig. 8. Visual results of small objects on DOTA-v1.0 dataset. The red box is a missed detection box, while the orange box is a false detection box.

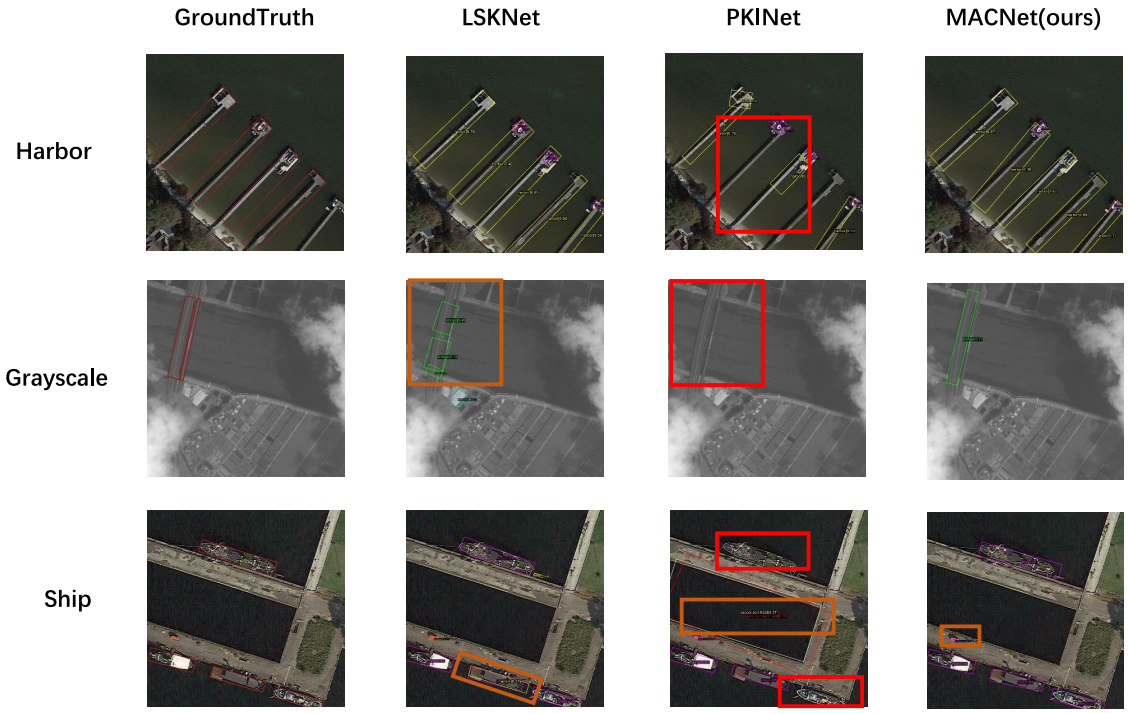


Fig. 9. Visual results of HAR objects on DOTA-v1.0 dataset. The red box is a missed detection box, while the orange box is a false detection box.

V. Conclusion

Detecting small objects and HAR objects in remote sensing images remains a significant challenge due to complex backgrounds and variations in object orientation. Accurate detection of these objects is crucial for applications. In this article, we propose a MACNet to address the challenges of detecting small objects and HAR objects in remote sensing images. By integrating the LKDB module, DC module, and a DKFIoU loss function, MACNet effectively tackles critical issues, including the feature attenuation in small objects, the oriented identification and feature extraction of the slender-shaped objects, and large localization deviations caused by scale and orientation sensitivity. Extensive experiments on

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DOTA-v1.0, DIOR-R, and HRSC2016 datasets demonstrate the superiority of MACNet over numerous methods. Specifically, MACNet achieves 81.82% mAP on small objects and 89.24% mAP on HAR objects in DOTA-v1.0, significantly outperforming baseline models. These results underline MACNet's potential to advance the field of remote sensing and improve object detection capabilities in challenging scenarios. While MACNet improves the detection of small objects, it might still struggle with objects with only a few pixels. In the future, we will explore the integration of super-resolution techniques to deal with the limited resolution of the remote sensing images.

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